CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

6.2 Elastic & Plastic Behaviour

Syllabus Section: 6. Deformation of Solids **Paper Type:** Structured Questions (Paper 2) **Total:** 10 Questions (110 Marks)

Question 1 Easy [9 marks]

Part (a) [2 marks]

Fig. 1.1 shows an incomplete extension–load graph obtained by adding multiple 1.0 N weights to a spring and recording the extension.

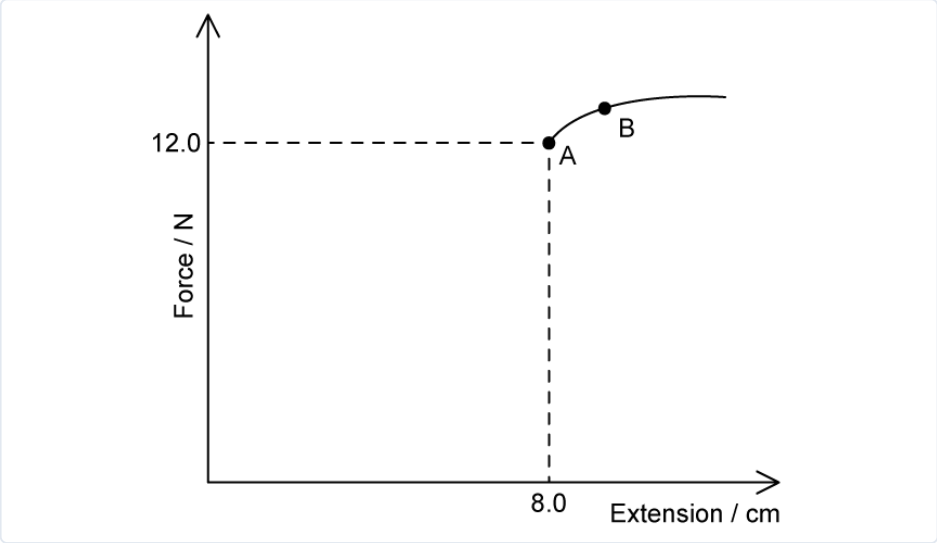


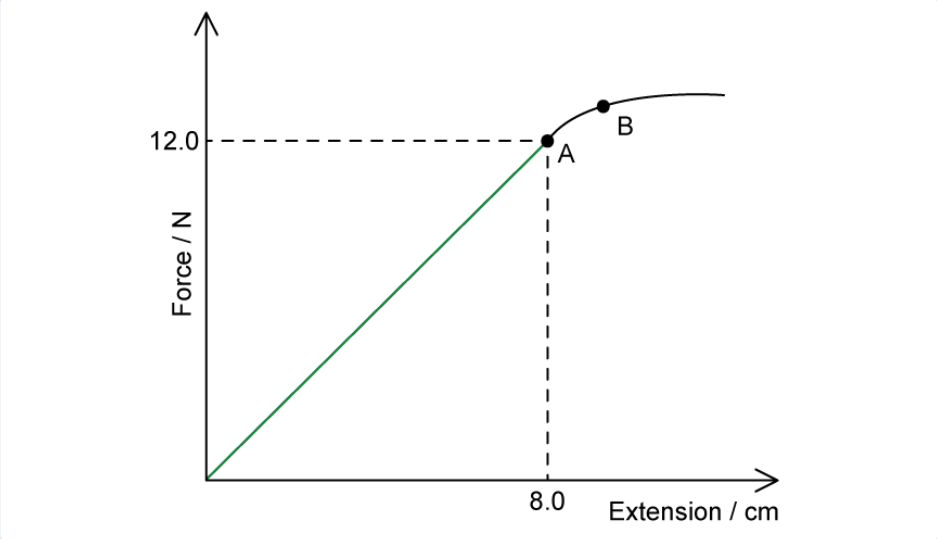
Fig. 1.1

The spring obeys Hooke's law up to point A.

Draw a suitable line on Fig. 1.1 to show the relationship between force and extension up to point A.

Mark Scheme (2 marks):

A graph that would score 2 marks should look like:



- Straight line starting at the origin (0,0); *[1 mark]*
- Ending at point A; *[1 mark]*

[Total: 2 marks]

We are told that the spring obeys Hooke's law up to point A. This means that the extension of the spring will be ***directly proportional*** *to the applied load. All directly proportional relationships will result in a graph which is a* ***straight line*** *through the* ***origin****.*

Part (b) **[1 mark]**

State the relevance of point A.

Mark Scheme (1 marks):

Point A is:

- Limit of proportionality; *[1 mark]*

[Total: 1 mark]

A common mistake is to mix up the limit of proportionality and the elastic limit. The limit of proportionality is the last point on the line where the load and extension are directly proportional.

Part (c) **[3 marks]**

Use Fig. 1.1 to calculate the energy stored in the spring when it is extended 8.0 cm. Show your working clearly.

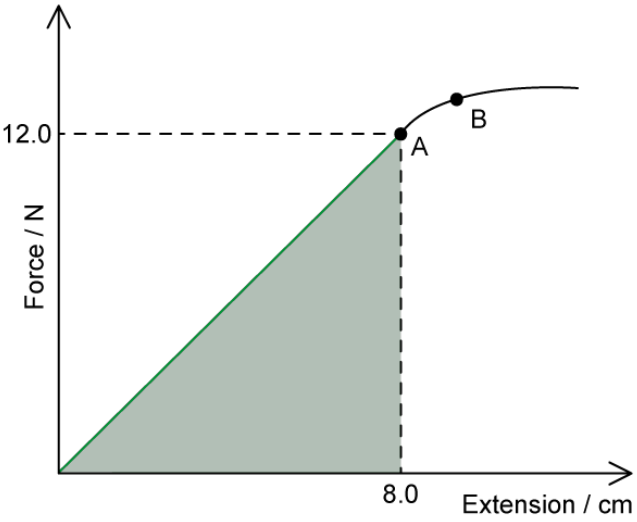
Mark Scheme (3 marks):

List the known quantities:

- Force applied to the spring, F = 12.0 N (from Fig. 1.1)
- Extension of the spring, ΔL = 8.0 cm = 0.08 m (from Fig. 1.1)

Calculate area under the graph:

- Energy stored in the spring, E = Area under graph = Area of a right-angled triangle; *[1 mark]*



- Area of a right-angled triangle = $\frac{1}{2}$ base × height

Substitute in values:

- $E = \frac{1}{2} \times 12.0 \times 0.08$ *[1 mark]*
- E = 0.48 J *[1 mark]*

[Total: 3 marks]

Part (d) **[3 marks]**

Fig. 1.1 shows the spring exhibits *elastic deformation* up to point B.

(i) State the name given to point B.

[1]

(ii) Explain what is meant by elastic deformation.

[2]

Mark Scheme (3 marks):

(i) Point B is:

- The elastic limit; *[1 mark]*

(ii) Elastic deformation is:

- When the load extending the spring is removed; *[1 mark]*
- The spring will return to its original length; *[1 mark]*

[Total: 3 marks]

Question 2 Easy [10 marks]

Part (a) [2 marks]

Define the term plastic deformation.

Mark Scheme (2 marks):

Plastic deformation is:

- When the load is removed; *[1 mark]*
- The object will not return to its original shape or length; *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

A student applies a range of forces to a metal wire and records the extension. The force–extension graph of his results is shown Fig. 1.1.

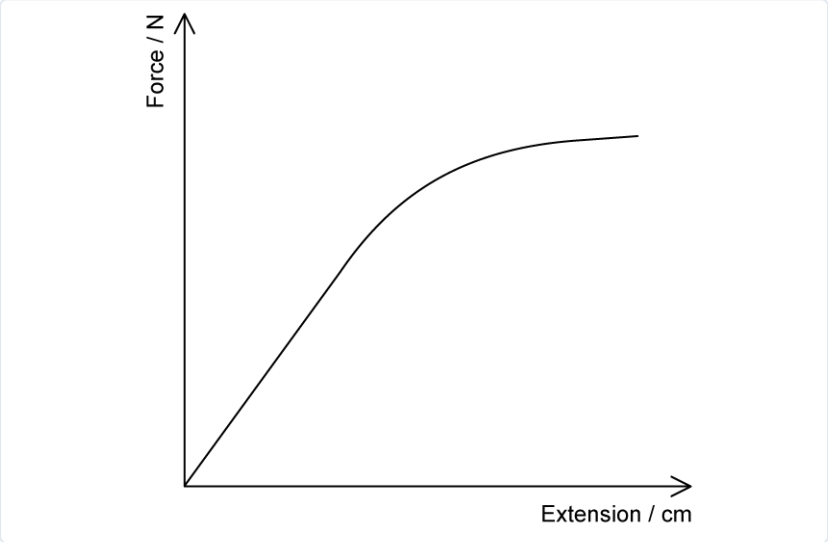


Fig. 1.1

(i) Mark the elastic limit on Fig. 1.1. Label this E.

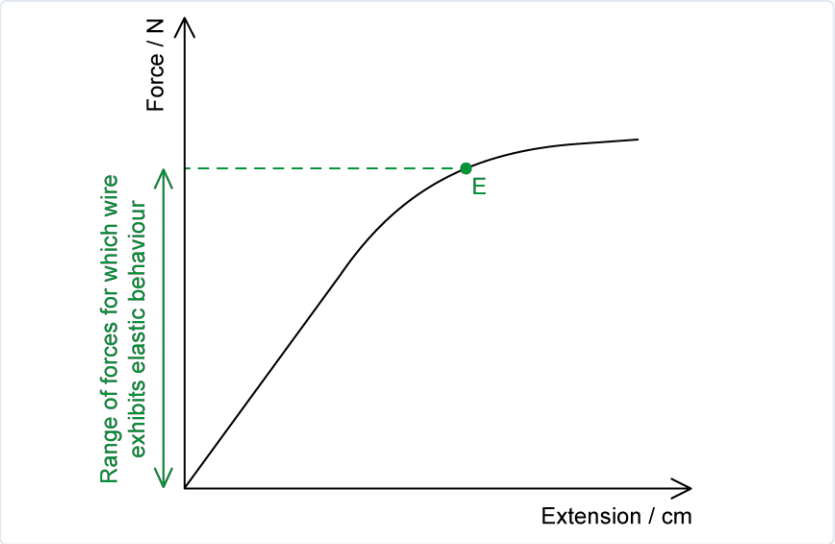
[1]

(ii) Show clearly on the y–axis of Fig. 1.1 the range of forces which cause the wire to exhibit elastic behaviour.

[1]

Mark Scheme (2 marks):

A graph that would score 2 marks should look like:



(i) Mark the elastic limit on Fig. 1.1:

- Point labelled E just after the straight line section of the graph; *[1 mark]*

(ii) Show clearly on the *y*–axis of Fig. 1.1 the range of forces which cause the wire to exhibit elastic behaviour:

- Forces from 0 N up to point in line with E marked on y axis; *[1 mark]*

[Total: 2 marks]

*Remember that the elastic limit lies on the **curved** part of the graph and is always **above** the limit of proportionality. The limit of proportionality is the last point on the straight line section of the graph.*

Part (c) **[3 marks]**

After the student has applied 15.0 N to the wire he reduces the force applied to it in small increments and records the extension.

On Fig. 1.2, draw a line to show how the relationship between the force acting on the wire and the extension as all the force is reduced.

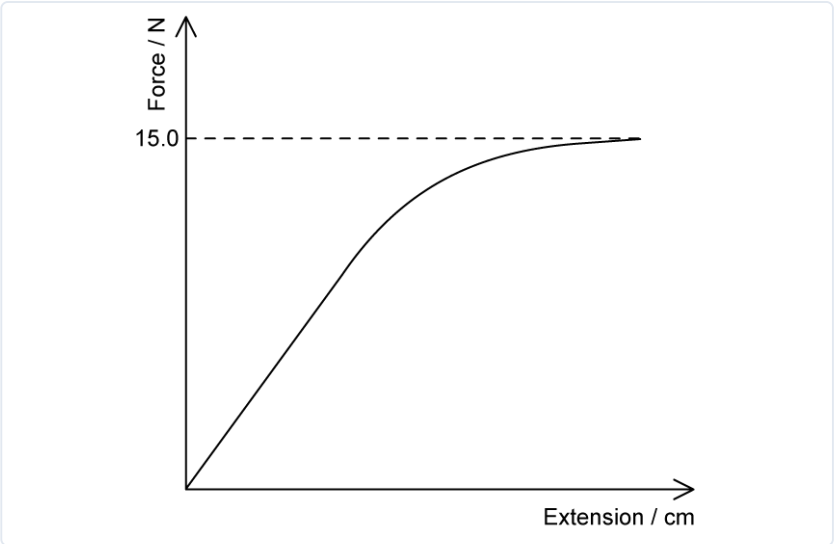
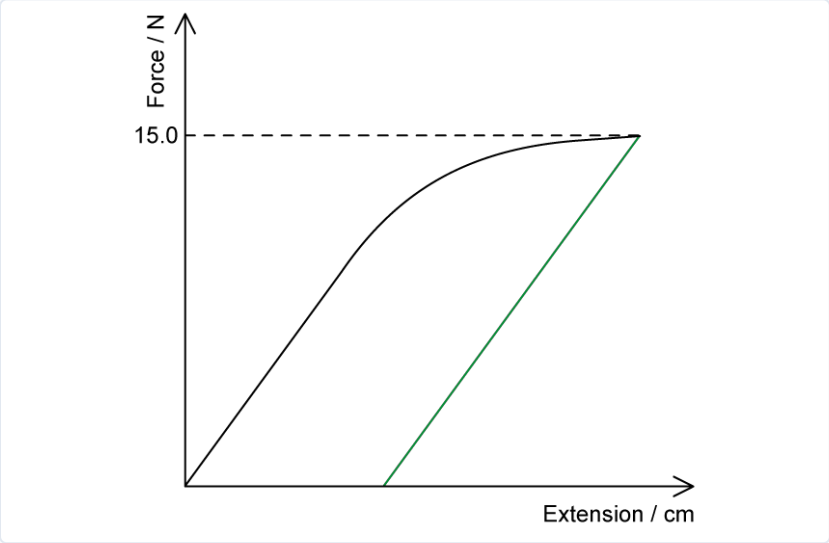


Fig. 1.2

Mark Scheme (3 marks):

A graph that would score 3 marks should look like:



- Straight line; *[1 mark]*
- From end of graph to a positive value on the x–axis; *[1 mark]*
- With the same gradient as the straight part of given graph; *[1 mark]*

[Total: 3 marks]

When a force of 15.0 N is applied, we can see that the curve is a good distance beyond the straight section of the graph, therefore the wire will have undergone ***plastic deformation****.*

Part (d) [3 marks]

Another identical spring is loaded with weights.

It has an extension of 8.0 cm and stores 24 J of elastic potential energy.

Calculate the spring constant of this spring.

Mark Scheme (3 marks):

Calculate the spring constant of the spring:

List the known quantities:

- Extension, x = 8.0 cm = 0.08 m
- Elastic potential energy, E_P = 24 J

Recall the equation for elastic potential energy:

- $E_P = \frac{1}{2}kx^2$ *[1 mark]*

Rearrange for *k* :

- $k = \frac{2E_P}{x^2} = \frac{2 \times 24}{0.08^2}$ *[1 mark]*
- $k = 7500 \text{ N m}^{-1}$ *[1 mark]*

[Total: 3 marks]

Question 3 Easy [7 marks]

Part (a) [2 marks]

Define

(i) stress

[1]

(ii) strain.

[1]

Mark Scheme (2 marks):

(i) Stress is defined as:

- Force / load applied per unit cross-sectional area *[1 mark]*

(ii) Strain is defined as:

- The ratio of extension to original length *[1 mark]*

Part (b) [2 marks]

A student performs an experiment to determine the Young modulus of a metal wire.

State **two** measurements, other than the applied load, that are required to determine the Young modulus of the metal.

For each measurement, name a suitable measuring instrument.

Mark Scheme (2 marks):

Measurements required to determine the Young modulus of the metal:

- Measurement: Original length (of wire)

  AND

  Instrument: Metre rule **OR** tape measure *[1 mark]*

- Measurement: Diameter (of wire)

  AND

  Instrument: Micrometer (screw gauge) **OR** (digital) callipers *[1 mark]*

Answers must link the dimension to the correct instrument, e.g. measuring length with a micrometer would not be reasonable.

Part (c) [3 marks]

A force F is applied to a uniform wire. The variation with extension x of F is shown in Fig. 4.1.

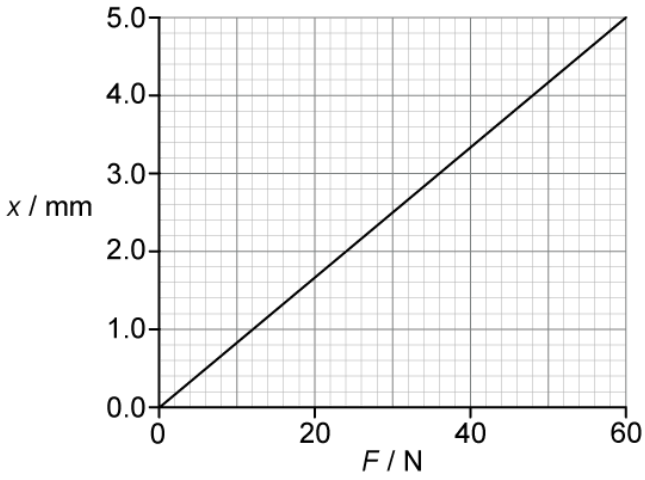


Fig. 4.1

The wire obeys Hooke's law up to a force of 60 N.

(i) Calculate the elastic potential energy stored in the wire when F = 60 N.

[2]

(ii) A force of greater than 60 N is now applied to the wire.

Describe how it may be checked that the elastic limit of the wire has not been exceeded.

[1]

Mark Scheme (3 marks):

(i) To calculate the elastic potential energy stored in the wire when F = 60 N:

- Recall how to determine elastic potential energy from the force-extension graph:

E = area under the graph

OR

$E = \frac{1}{2}Fx$ *[1 mark]*

- Calculate the elastic potential energy:

From the graph: when $F = 60$ N, $x = 5.0$ mm $= 0.005$ m

$E = \frac{1}{2} \times 60 \times 0.005$

$E = 0.15$ J *[1 mark]*

(ii) To check that the elastic limit of the wire has not been exceeded:

- Remove the force / F, and the wire should return to its original length *[1 mark]*

Question 4 Medium [12 marks]

Part (a) [8 marks]

Fig. 1.1 is a graph of stress against strain for two wires, **X** and **Y**. The wires are made from different materials but have the same dimensions.

(i) Describe, giving reasons for your answer, three properties of material **X**.

[6]

(ii) State the meaning of points F_x and F_y and explain their significance.

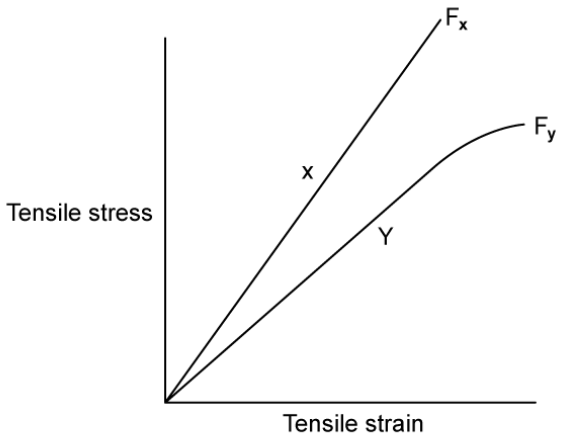


Fig. 1.1

[2]

Mark Scheme (8 marks):

(i) Describe, giving reasons for your answer, three properties of material **X**:

*Any **three** pairs (reason mark is dependent on being given for the correct description):*

- Larger Young Modulus / is stiffer than material Y; *[1 mark]*
  - *Reason*: the gradient is steeper; *[1 mark]*
- Does not undergo (any) plastic deformation before breaking; *[1 mark]*
  - *Reason*: the line does not curve / is straight to the end; *[1 mark]*
- Obeys Hooke's Law; *[1 mark]*
  - *Reason*: stress is always proportional to strain; *[1 mark]*
- Stronger (than Y); *[1 mark]*
  - *Reason*: higher breaking stress / needs greater force per area to break; *[1 mark]*
- Brittle; *[1 mark]*

- *Reason*: does not exhibit plastic behaviour; *[1 mark]*

- Stores less elastic strain energy (for given values of strain); *[1 mark]*
  - *Reason*: extension is smaller / area under graph is always less; *[1 mark]*

(ii) State the meaning of points F_x and F_y and explain their significance:

- Breaking point / ultimate tensile strength; *[1 mark]*
- Shows the maximum stress per strain which material can withstand / how strong the material is; *[1 mark]*

[Total: 8 marks]

Dependent marks are given (or withheld) when an explanation is only worthy of credit if it attached to an explanation of something happening. Answers where you are asked to describe something (such as a graph changing, a spring stretching or a ball bouncing) in terms of a law of physics or an equation, will often have dependent marks.

Make sure you give complete answers, rather than statements which are too short. Remember, your teacher may know you know something and be used to you skipping parts of answers, but your examiner has nothing to go on but what you write in the exam!

Part (b) [2 marks]

State and explain which wire would be more suitable for use as cables and structural beams.

Mark Scheme (2 marks):

State and explain which wire would be more suitable for use as cables and structural beams:

- Wire **Y**; *[1 mark]*
- Because cables and structural beams need to be able to bend / deform (by some amount) before they break; *[1 mark]*

[Total: 2 marks]

Part (c) [2 marks]

A group of physics students are told that they will be performing the investigation which leads to the graph in part (a).

State two safety precautions which they must take and explain your answer.

Mark Scheme (2 marks):

State two safety precautions which they must take and explain your answer:

- Wear safety glasses / goggles, <u>because</u> the wires will snap (and may cause eye damage); *[1 mark]*

- Wear safety shoes / steel-toed boots / arrange a safety mat under the weights, because (when the wire snaps) the weights will fall; *[1 mark]*

[Total: 2 marks]

Question 5 Medium [15 marks]

Part (a) [4 marks]

Fitness equipment often uses springs as a method of creating resistance. One example is the Push Down Bar, which the manufacturer says can help a person who uses it to 'increase strength and burn calories'.

To use a Push Down Bar a person applies force by pushing the handles towards each other as shown in Fig. 1.1.



Fig. 1.1

Fig. 1.2 shows the construction of the Push Down Bar.

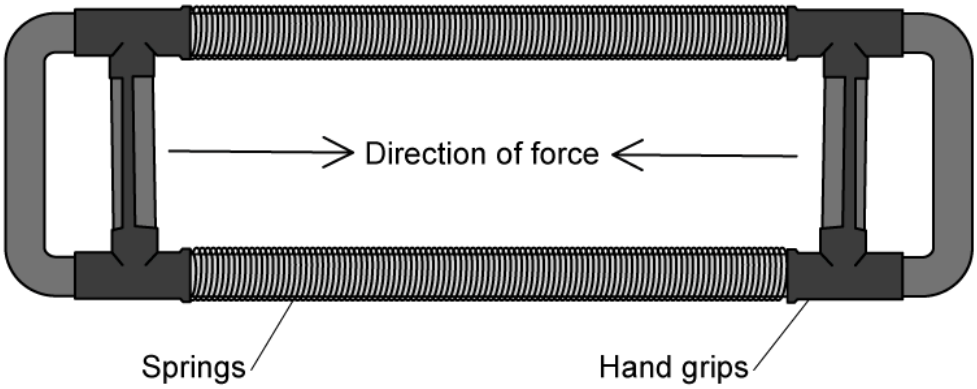


Fig. 1.2

(i) State the type of force used when exercising with this equipment.

[1]

(ii) Explain why using this equipment would help the person burn calories.

[3]

Mark Scheme (4 marks):

(i) State the type of force used when exercising with this machine;

- Compressive (force); *[1 mark]*

(ii) Explain why using this machine would help the person to burn calories:

- Force is applied across a distance; *[1 mark]*
- Work done to compress springs; *[1 mark]*
- Energy transferred from chemical energy store of the person exercising; *[1 mark]*

[Total: 4 marks]

Part (b) **[5 marks]**

The relationship between applied force and change in length of the springs was measured for a range of values of force.

The results are plotted on the graph in Fig. 1.3.

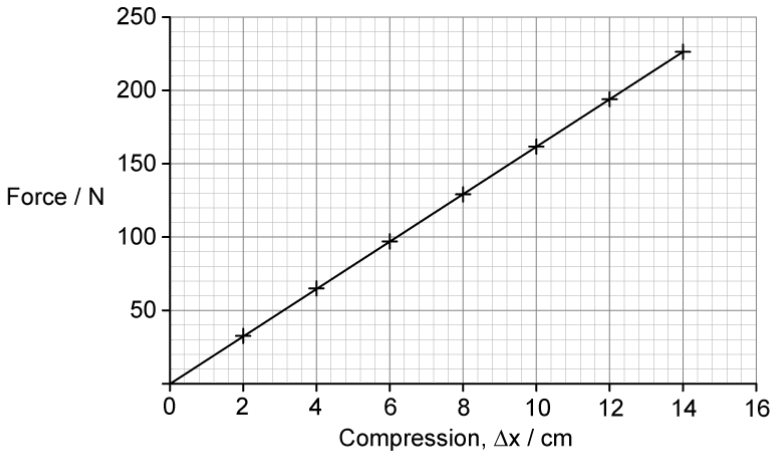


Fig. 1.3

(i) State, with a reason, whether the spring obeys Hooke's law over the range of values tested.

[2]

(ii) Use the graph in Fig. 1.3 to calculate the spring constant, stating an appropriate unit.

[3]

Mark Scheme (5 marks):

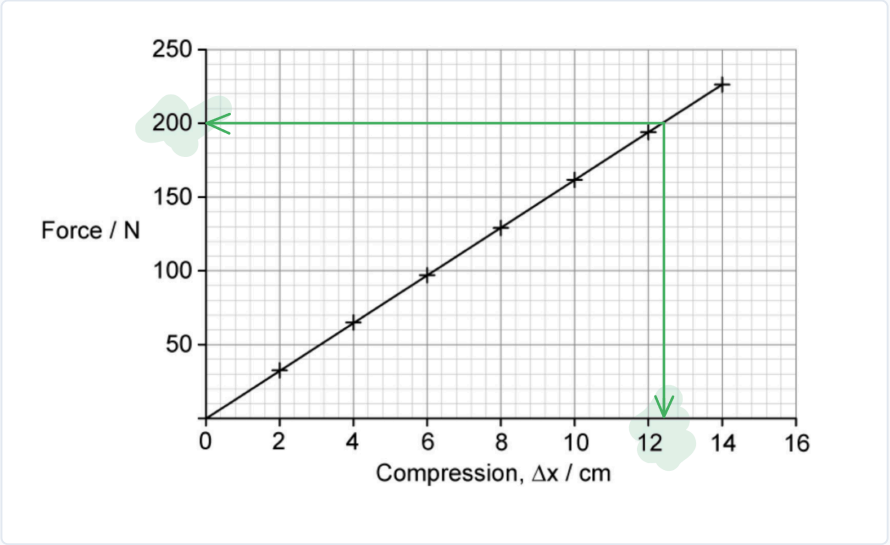
The force exerted by the spring over a range of compressions was measured and a graph plotted:

(i) State, with a reason, whether the spring obeys Hooke's law over the range of values tested:

- Yes, the spring obeys Hooke's law; *[1 mark]*
- (Because) the graph is a straight line through the origin; *[1 mark]*

(ii) Use the graph in Fig. 1.3 to calculate the spring constant, stating an appropriate unit:

- Clear attempt marked on the graph to find the gradient *[1 mark]*



- Gradient, $k = \frac{200 \text{ N}}{0.124 \text{ m}}$ (values must be separated by at least two-thirds of the line) *[1 mark]*
- $k = 1600 \text{ Nm}^{-1}$ *[1 mark]*

[Total: 5 marks]

In graphical questions the examiner is often looking for evidence that you know how to use a graph to analyse data.

*Make sure that you **write on the graph**, showing the points where you are taking the gradient from, or shading or boxing an area you are going to calculate.*

This way, even if the working is incorrect (and scores no marks) you will get a 'method mark' for demonstrating that you understood the principle.

Part (c) **[4 marks]**

Derive the formula for the elastic potential energy stored by the spring from the graph of force against extension.

Mark Scheme (4 marks):

Derive the formula for the elastic potential energy stored by the spring from the graph of force against extension:

Work done

- The work done on / by the spring to stretch or compress it is equal to the elastic potential energy stored in the spring; *[1 mark]*
- Work done, $\Delta W = F \times d$ (by definition)
- *d*, distance moved in the direction of the force = Δ*x*, (the compression or extension)

Area under the force-compression graph

- Therefore, the area under the graph represents the work done / energy required to compress / extend the spring; *[1 mark]*
- Area under graph = area of a right-angle triangle = ½ × base height = ½ × Δ*x* × *F [1 mark]*
- Which is the same as elastic potential energy $E_p = \frac{1}{2}Fx$ *[1 mark]*

[Total: 4 marks]

Part (d) **[2 marks]**

State and explain whether the material chosen for the spring for the equipment in Fig. 1.1 should exhibit elastic or plastic behaviour.

Mark Scheme (2 marks):

State and explain whether the material chosen for the spring for the equipment in Fig. 1.1 should exhibit elastic or plastic behaviour.

- Elastic; *[1 mark]*
- The spring should return to original length / shape once the applied force is removed; *[1 mark]*

[Total: 2 marks]

Question 6 Medium [12 marks]

Part (a) [3 marks]

A manufacturer produces springs for use in school laboratory investigations. As part of quality control the springs are spot-checked by measuring the extension produced for certain applied loads.

The graph of the testing data is shown in Fig. 1.1

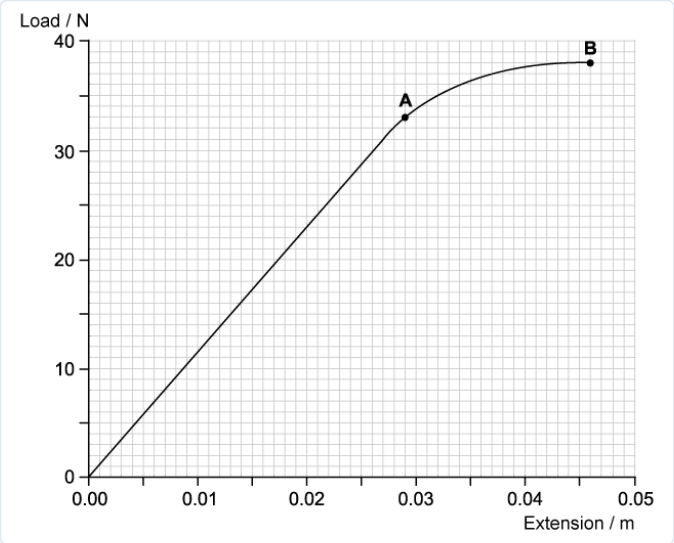


Fig. 1.1

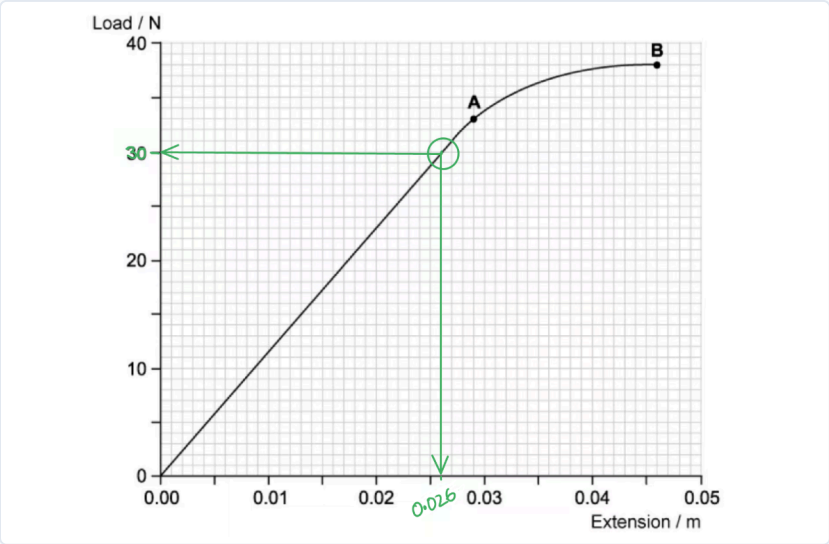
Use a graphical method to calculate the spring constant of the spring.

Show your working.

Mark Scheme (3 marks):

To calculate the spring constant of the spring:

Find the gradient of the graph:



- A large triangle (must occupy at least half of the line) or other method clearly marked, on the straight part of the line; *[1 mark]*
- Spring constant = gradient, therefore $k = \frac{\Delta y}{\Delta x} = \frac{30}{0.026}$ *[1 mark]*
- $k = 1154 \text{ Nm}^{-1}$ *[1 mark]*

[Total: 3 marks]

*Notice that the first mark here is for taking the gradient of the **straight** part of the graph, using **as much of the line as possible**, and **marking it clearly** on the graph.*

*If you mark out a large section of line but get the gradient wrong, you **will** still get this mark.*

*If you use a small triangle, even if you are lucky enough to get the gradient correct, you will **not** get this mark.*

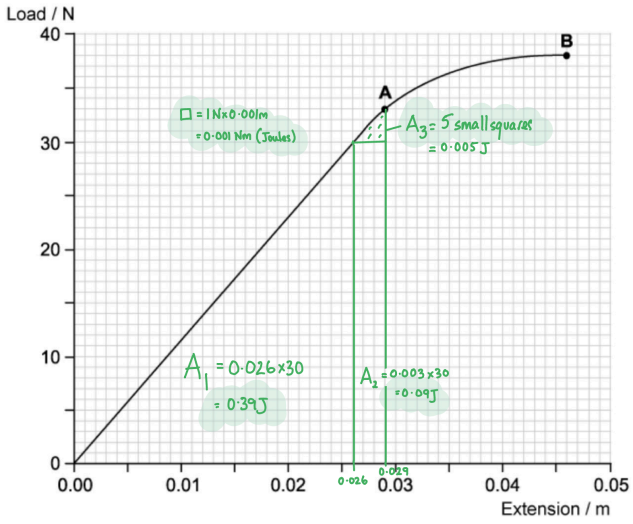
Part (b) **[3 marks]**

Show that the work done in extending the spring up to point **A** is approximately 0.5 J.

Mark Scheme (3 marks):

Show that the work done in extending the spring up to point **A** is approximately 0.5 J:

Determine the area under the graph by dividing it into sections:



*Any **two** from:*

- Area of triangle, $A_1 = \frac{1}{2} \times 0.026 \times 30 = 0.39$ J *[1 mark]*
- Area of rectangle, $A_2 = 0.003 \times 30 = 0.09$ J *[1 mark]*
- Area of small square, $A_{\square} = 1 \times 0.001 = 0.001$ *[1 mark]*
- Count small squares = 5 *[1 mark]*
- So, A_3 = 0.005

Add up all *A* values for total work done:

- $W = 0.39 + 0.09 + 0.005 = 0.485$ J *[1 mark]*

[Total: 3 marks]

The exact values you get for splitting the area under the graph into triangles and rectangles, and then counting squares, will depend on where you judged the sections to be.

The examiner would check your working and mark it correct for your method. If you want to check your method, first check if you got the correct area. If so, then you did well. If not, ask a friend or teacher to take a look. The most likely mistake is in getting the decimal point in the correct place for the values on the x-axis.

The final mark checks the accuracy of your method overall.

Part (c) [3 marks]

When the spring reaches an extension of 0.046 m, the load on it is gradually reduced to zero.

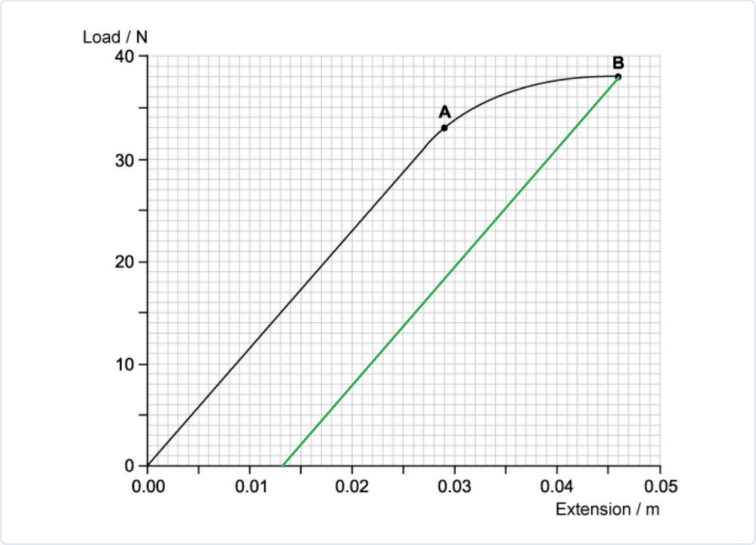
On the graph in Figure 1.1, sketch how the extension of the spring will vary with load as the load is reduced to zero.

Explain why the graph has this shape.

Mark Scheme (3 marks):

Sketch the extension of the spring as the load is reduced to zero and explain the graph's shape:

*A graph that would score **2 marks** looks like:*



- Straight line from point **B** to the *x*-axis; *[1 mark]*
- Ends on *x*-axis in range 0.013 m - 0.014 m; *[1 mark]*

Explanation

- (The spring does not return to its original length because) it has been permanently deformed / has undergone plastic deformation; *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

Without further calculation, compare the total work done by the spring when the load is removed with the work that was done by the load in producing the extension of 0.046 m.

Explain how this is represented on the graph drawn in part (c).

Mark Scheme (3 marks):

Compare the total work done unloading and loading the spring and explain how this is represented on the graph:

- The total work done when the load is removed / work done by spring is less than the work done by the load (in producing the extension); *[1 mark]*
- The area under a force-extension graph is the work done; *[1 mark]*
- Therefore as the load is reduced to zero the area under the unloading curve is smaller since less work is done (OR reverse argument); *[1 mark]*

[Total: 3 marks]

Question 7 Medium [11 marks]

Part (a) [2 marks]

Materials scientists are asked to produce a material which would be suitable to use in constructing the wing of an aeroplane.

The team of scientists work with two materials, an aluminium alloy and a carbon fibre composite.

Suggest a material property which the scientists are likely to investigate and report on, and give a reason.

Mark Scheme (2 marks):

A property which the materials scientists are likely to test for:

*Any one **pair** from:*

- Breaking stress; *[1 mark]*
  - *Reason*: to know how much force the material can withstand before breaking; *[1 mark]*
- Young Modulus; *[1 mark]*
  - *Reason*: to know the stiffness / how much material can deform before breaking; *[1 mark]*

[Total: 2 marks]

Part (b) [6 marks]

Fig. 1.1 shows the stress-strain graph that the scientists obtain for the aluminium alloy.

They have labelled a point **Q** on the graph.

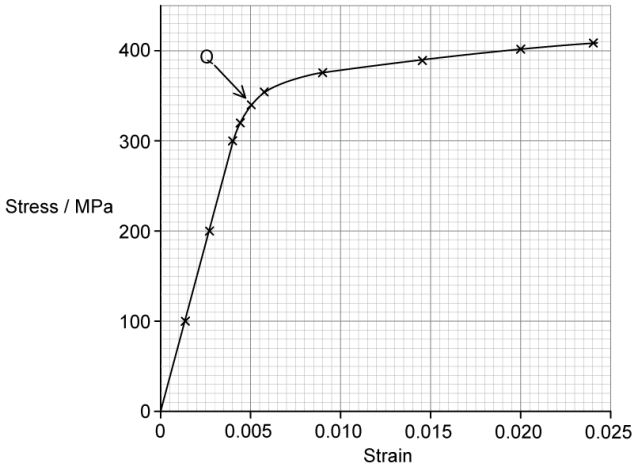


Fig. 1.1

(i) State the name of point **Q** and suggest why the scientists have drawn attention to it.

[3]

(ii) Use the graph to determine the Young modulus of the aluminium alloy. Show your working.

[3]

Mark Scheme (6 marks):

For the aluminium alloy:

(i) State the name of point **Q** and suggest why the scientists have drawn attention to it:

- Elastic limit: *[1 mark]*
- When more force is applied / beyond this point the material will no longer return to its original length / shape when the force is removed; *[1 mark]*
- Wings should deform / bend in use **AND** after must return to original shape; *[1 mark]*

(ii) Use the graph to determine the Young modulus of the aluminium alloy. Show your working:

- Clear attempt to find gradient of straight part of line marked on graph; *[1 mark]*

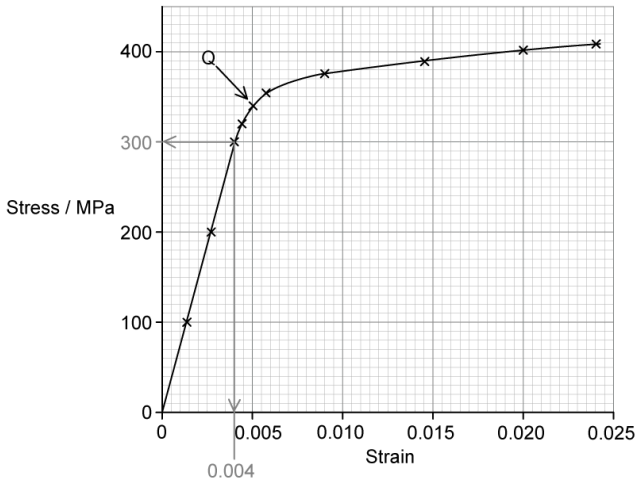


Fig. 1.1

Calculate the gradient:

- $gradient = \frac{\Delta y}{\Delta x} = \frac{300 \times 10^6}{0.004} = 7.5 \times 10^{10}$ *[1 mark]*

State the Young Modulus including units:

- $E = 7.5 \times 10^{10}$ Pa *[1 mark]*

[Total: 6 marks]

In graphical questions the examiner is usually looking for evidence that you know how to use a graph to analyse data.

*Notice that the first mark here is for taking the gradient of the **straight** part of the graph, using **as much of the line as possible**, and **marking it clearly** on the graph.*

*If you mark out a large section of line but get the gradient wrong, you **will still get this mark**.*

*If you use a small triangle, even if you are lucky enough to get the gradient correct, you will **not** get this mark.*

Part (c) **[3 marks]**

The aluminium alloy wire has a diameter of 1.2 mm. Calculate the elastic potential energy stored in the wire when it shows an extension of 7.5 cm under a total stress of 190 MPa.

Mark Scheme (3 marks):

Calculate the elastic potential energy stored in the wire:

List the known quantities:

- Stress, σ = 190 MPa = 190 × 10^6 Pa
- Extension, x = 7.5 cm = 0.075 m
- Diameter of the wire, d = 1.2 mm = 1.2 × 10^{-3} m

Calculate the cross-sectional area of the wire, A:

- $A = \pi r^2$
- $A = \pi \frac{1.2 \times 10^{-3}}{2}^2$
- $A = 1.13 \times 10^{-6}$ m^2 *[1 mark]*

Calculate the force, F applied to the wire:

- Stress, $\sigma = \frac{F}{A} \rightarrow F = \sigma A$
- $F = 190 \times 10^6 \times 1.13 \times 10^{-6}$
- $F = 214.7$ N *[1 mark]*

Calculate the elastic potential energy, E_p stored in the wire:

- $E_p = \frac{1}{2} F x$
- $E_p = \frac{1}{2} \times 214.7 \times 0.075$
- $E_p = 8.05$ J *[1 mark]*

[Total: 3 marks]

Question 8 Medium [7 marks]

Part (a) [2 marks]

A steel bar is 40 mm long and has cross-sectional area 4.5×10^{-4} m^2.

The bar is compressed using a vice so that the length is reduced by 0.20 mm.

The Young Modulus of steel = 2.1×10^{11} Pa.

Calculate the compressive force exerted on the bar.

Mark Scheme (2 marks):

Calculate the compressive force exerted on the bar:

List the known quantities:

- Cross-sectional area of bar, A = 4.5×10^{-4} m^2
- Length of bar, x = 40 mm
- Compression of bar, Δx = 0.20 mm
- Young-Modulus, E = 2.1×10^{11} Pa

Use the full equation for the Young Modulus:

- $E = \frac{F}{A} \times \frac{x}{\Delta x}$

Rearrange to make *F* the subject and calculate:

- $F = \frac{E \times A \times \Delta x}{x} = \frac{2.1 \times 10^{11} \times 4.5 \times 10^{-4} \times 0.2}{40}$ *[1 mark]*
- $F = 472 \quad 500$ N

Write the answer to two sig figs and include the unit:

- $F = 4.7 \times 10^{5}$ N *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

Calculate the work done compressing the bar.

Mark Scheme (2 marks):

Calculate the work done compressing the bar:

List the known quantities:

- Compressive force on bar, F = 4.7×10^{5} N (from part (a))
- Compression of bar, Δx = 0.2×10^{-3} m (from part (a))

Use the equation for elastic energy stored in the bar:

- $E_P = \frac{1}{2} \times F \times s$
- $E_P = \frac{1}{2} \ 4.7 \times 10^{5} \times 0.2 \times 10^{-3}$ *[1 mark]*
- $W = 47 \ \text{J}$ *[1 mark]*

[Total: 2 marks]

Part (c) [3 marks]

For the compression described in part (a)

(i) Sketch a graph to show the compression, assuming that the elastic limit has not been reached.

[2]

(ii) Suggest how the elastic potential energy could be ascertained from the graph.

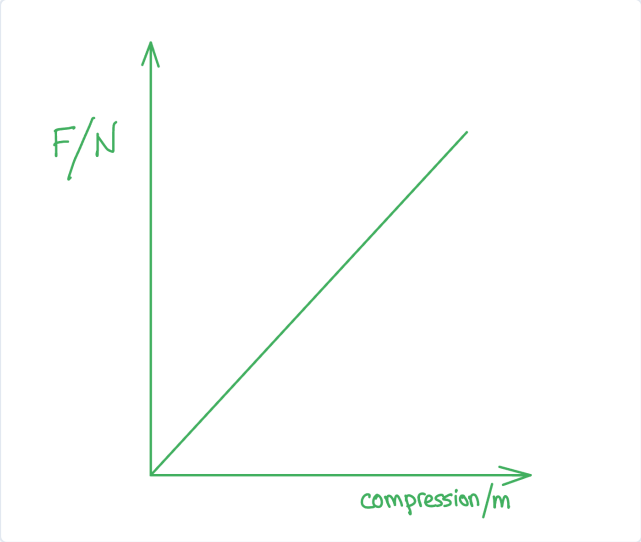
[1]

Mark Scheme (3 marks):

For the compression described in part (a)

(i) Sketch a graph to show the compression, assuming that the elastic limit has not been reached:

*A graph which would score **two marks** looks like:*



- Both axes labelled force and compression as shown; *[1 mark]*
- Straight line through the origin; *[1 mark]*

(ii) Suggest how the elastic potential energy could be ascertained from the graph:

- The area under the graph; *[1 mark]*

> ***[Total: 3 marks]***

Question 9 Hard [16 marks]

Part (a) [4 marks]

A light spring of length 2.2 cm is hung from a fixed point. An object of weight 4.0 N is hung from the other end of the spring. Fig. 1.1 shows the length of the spring when the object is in equilibrium.

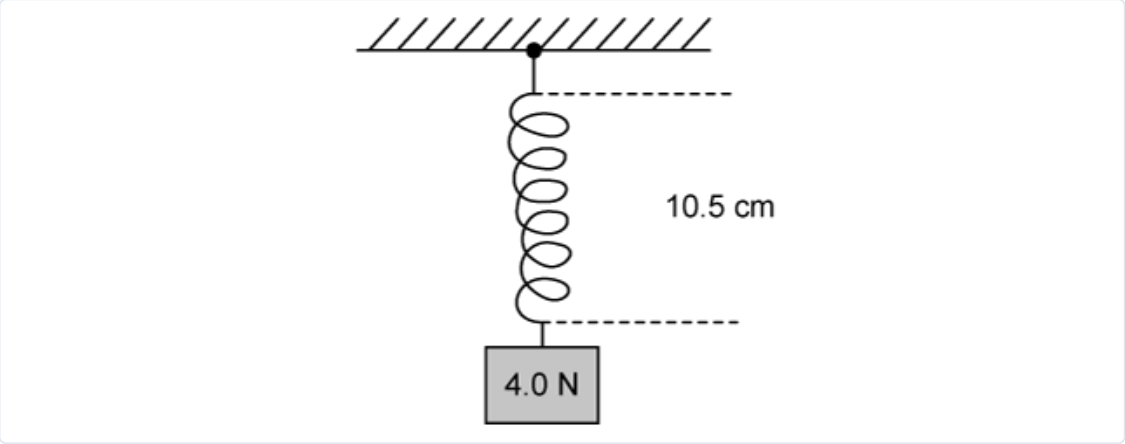


Fig. 1.1

The object is then pulled vertically downwards and is no longer in its equilibrium position. This is shown in Fig. 1.2.

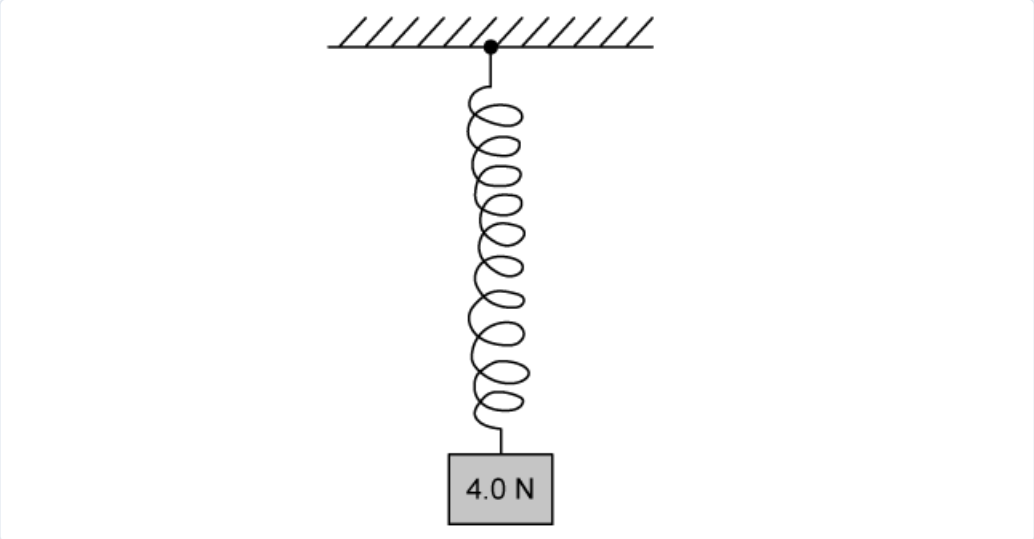


Fig. 1.2

The change in elastic potential energy ΔE between the spring in Fig. 1.1 and Fig. 1.2 is 0.30 J.

Calculate the length of the spring in Fig. 1.2.

Mark Scheme (4 marks):

List the known quantities:

- Change in elastic potential energy, ΔE = 0.30 J
- Force, F = 4.0 N
- Extension of spring in Fig. 1.1, Δx_a = 10.5 cm – 2.2 cm = 8.3 cm = 0.083 m
- Original length, L = 2.2 cm = 0.022 m

Calculate the spring constant of the spring, k:

- Hooke's law: $F = k \Delta x$
- $k = \frac{F}{\Delta x} = \frac{4.0}{0.083}$ = 48.19 = 48 N m^{-1} *[1 mark]*

Determine the equation for the change in elastic potential energy, ΔE:

- Elastic potential energy, $E = \frac{1}{2} \times k \times (\Delta x)^2$
  - It is the same spring, so both scenarios have the same value of k
- $\Delta E = \frac{1}{2} k ((\Delta x_b)^2 - (\Delta x_a)^2)$ *[1 mark]*
  - Where Δx_a is the extension of the spring in Fig. 1.1 and Δx_b is the extension in Fig. 1.2

Rearrange the equation for the extension in Fig. 1.2, Δx_b:

- $\Delta x_b = \sqrt{\frac{2\Delta E}{k} + \Delta x_a{}^2}$
- $\Delta x_b = \sqrt{\frac{2 \times 0.30}{48} + 0.083^2}$ = 0.139 = 0.14 m *[1 mark]*

Calculate length of the spring:

- Length = $L + \Delta x_b$ = 0.022 + 0.14 = 0.162 m (16.2 cm) *[1 mark]*

[Total: 4 marks]

*Remember to convert to **metres** from **centimetres** for all the extension and lengths. This is because the change in elastic potential energy, E is measured in **Joules (J)**. The SI base unit for Joules is **N m**. Therefore, the calculation will not work if you keep all your values in cm!*

Part (b) **[3 marks]**

The object is released from its position shown in Fig. 1.2.

Calculate its initial acceleration.

Mark Scheme (3 marks):

List the known quantities:

- Spring constant, k = 48 N m^{-1} (from part a)
- Extension, Δx_b = 0.139 m (from part a)
- Weight, W = 4.0 N

Calculate the mass, *m*:

- *W* = *mg*
- $m = \frac{W}{g} = \frac{4.0}{9.81} = 0.408$ kg *[1 mark]*

Calculate the resultant force, *F* on the object:

- Downwards force, F_d = *W* = 4.0 N
- Upwards force, $F_u = k\Delta x_b$ = 48 × 0.139 = 6.672 N
- Resultant force = F_u - F_d = 6.672 – 4.0 = 2.672 N *[1 mark]*

Calculate acceleration, *a*:

- Resultant force, *F* = *ma*
- $a = \frac{F}{m} = \frac{2.672}{0.408} = 6.55$ m s^{-2} *[1 mark]*

[Total: 3 marks]

It is important to understand the direction of forces to understand this question. There isn't just the weight force pushing the spring downwards, but since the spring has been stretched when released it will go back to its equilibrium position. This is the upwards force, therefore the resultant force is upwards.

Part (c) **[6 marks]**

Fig. 1.3 shows a force–extension graph for the spring.

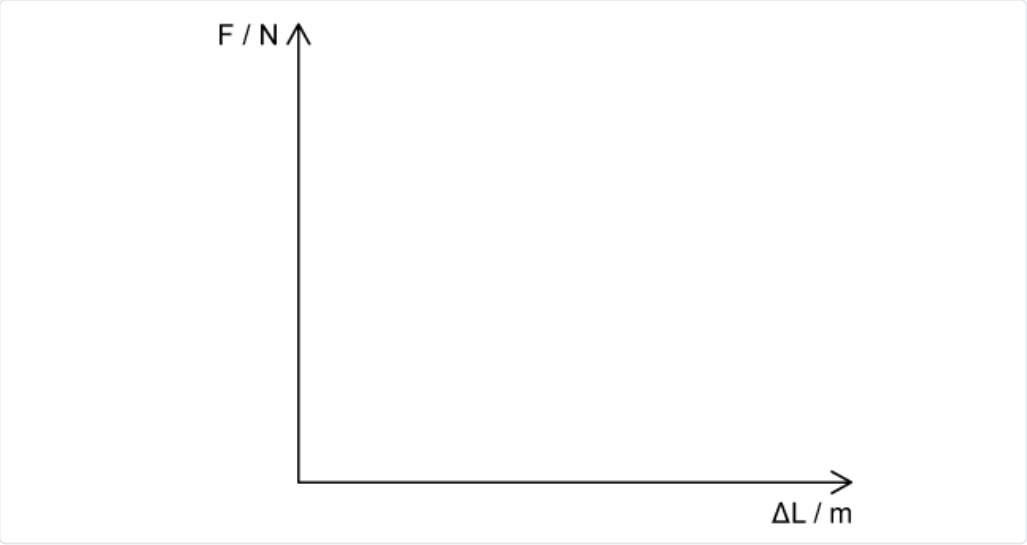


Fig. 1.3

Assuming that the spring is not extended beyond its limit of proportionality:

(i) Sketch a graph on Fig. 1.3 to show the key values from the spring in equilibrium (in Fig. 1.1) and when the spring is extended (in Fig. 1.2).

[4]

(ii) Show that the graph also demonstrates that the change in elastic potential energy stored, Δ*E* is 0.30 J.

[2]

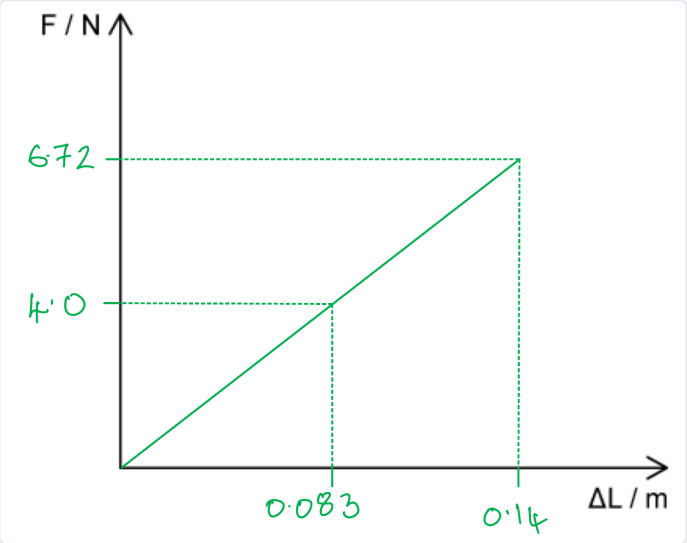
Mark Scheme (6 marks):

List the known quantities:

- Change in elastic strain energy, Δ*E* = 0.30 J
- Force on spring in Fig. 1.1 = 4.0 N
- Extension of spring in Fig. 1.1, Δx_a = 0.083 m
- Force on spring in Fig. 1.2 = 6.72 N (from part b)
- Extension of spring in Fig. 1.2, Δx_b = 0.14 m (from part a)

(i) Sketch a graph on Fig. 1.3 to show the key values from the spring in equilibrium

A graph that would score 3 marks should look like:



- Straight diagonal line upwards; *[1 mark]*
- Force and extension for spring in Fig. 1.1 correctly labelled as 4.0 N; *[1 mark]*
- Force and extension for spring in Fig. 1.2 correctly labelled as 6.72 N; *[1 mark]*
- The graph doesn't extend past (0.14, 6.72); *[1 mark]*

(ii) Show that the graph also demonstrates that the change in elastic potential energy stored is 0.30 J

Calculate change in elastic potential energy from graph, Δ*E*:

- Elastic potential energy stored = Area under the graph $= \frac{1}{2} F\,\Delta x$ *[1 mark]*
- Area under the graph for spring in Fig. 1.1 = 0.5 × 4.0 × 0.083 = 0.166 J
- Area under the graph for spring in Fig. 1.2 = 0.5 × 6.72 × 0.14 = 0.4704 J

- ΔE = 0.4704 – 0.166 = 0.3044 = 0.30 J *[1 mark]*

[Total: 6 marks]

The final mark is for not extending the graph further than the values you have calculated. This is because past the point of extension in Fig. 1.2, it is not known whether or not the spring still obeys Hooke's law. Therefore, the graph could still be a straight line ***or*** *could start to curve, but this is not known in the question.*

The area under the graph is a ***right–angled triangle****.*

The area is equal to ½ × base × height.

Part (d) **[3 marks]**

Fig. 3.1 now shows two different springs S_1 and S_2 . The top ends of the springs are attached to a rod.

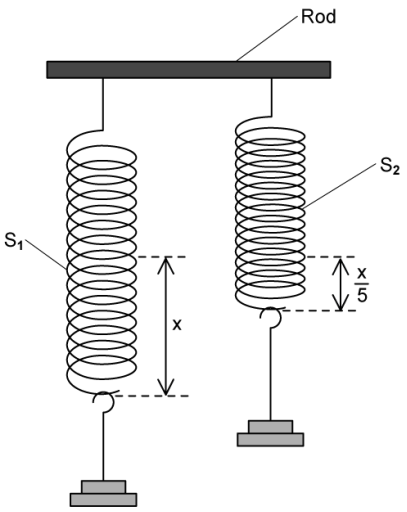


Fig. 3.1

A mass is hung from the bottom end of S_1 . The extension of S_1 is *x* and the elastic potential energy in the spring is 19 mJ. The same mass is hung from the bottom end of S_2 . The extension of S_2 is $\frac{x}{5}$ and its spring constant is 5 times that of S_1 .

Calculate the elastic potential energy in S_2 .

Mark Scheme (3 marks):

List the known quantities:

- Elastic potential energy in S_1 , E_1 = 19 mJ = 19 × 10^{-3} J
- Extension of S_1 = *x*
- Extension of $S_2 = \frac{x}{5}$
- Spring constant of S_1 = k_1
- Spring constant of S_2 , k_2 = $5k_1$

Determine an expression for the elastic potential energy of $S_2\,(E_2)$ in terms of that of $S_1\,(E_1)$:

- $E_1 = \frac{1}{2}k_1 x^2 = 19 \times 10^{-3}$ J
- $E_2 = \frac{1}{2}k_2 \frac{x}{5}^2 = \frac{1}{2}k_2 \frac{x^2}{25} = \frac{1}{2}5k_1 \frac{x^2}{25} = \frac{1}{2}k_1 \frac{x^2}{5}$ *[1 mark]*
- $E_2 = \frac{E_1}{5}$ *[1 mark]*

Calculate E_2:

- $E_2 = \frac{19 \times 10^{-3}}{5} = 3.8$ mJ *[1 mark]*

[Total: 3 marks]

Question 10 Hard [11 marks]

Part (a) [3 marks]

An adventure experience is shown in Fig. 1.1 below.

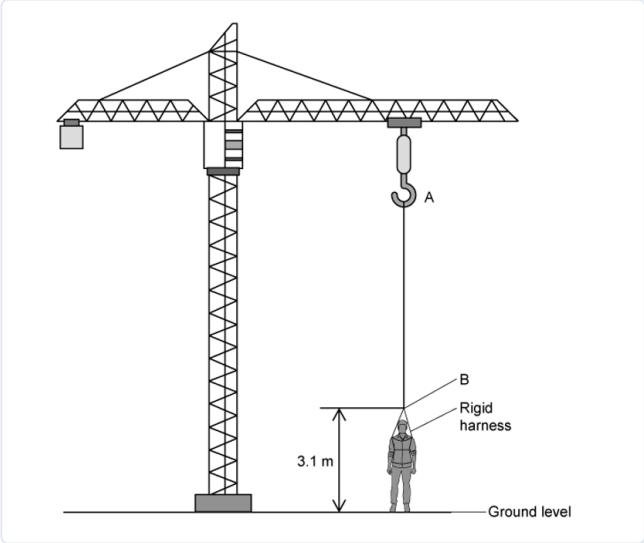


Fig. 1.1

The rider is strapped into a rigid harness attached to one end of an elastic rope AB. The rider and the rope behave in the same way as a mass–spring system.

The rope has an unstretched length of 27 m. When stretched, the rope obeys Hooke's law. The rider and harness have a total mass of 65 kg.

The rider is initially held at rest at ground level. The lower end of rope B is attached to the rigid harness at a point which is 3.1 m above the ground, so the harness doesn't injure the rider by being too close. The top end of the rope, A, is adjusted so the rope becomes unstretched when the rider is at the highest point of the ride. The rider is released and moves upwards, reaching a maximum height when the rope is at its unstretched (natural) length. The rider then oscillates vertically.

The height of point A above the ground is 51 m.

Calculate the spring constant of the rope neglecting air resistance and ignoring the mass of the rope in this question.

Mark Scheme (3 marks):

List the known quantities:

- Unstretched length, L = 27 m
- Height of B above the ground, x = 3.1 m
- Mass of rider and harness, m = 65 kg
- Height of A above the ground, h = 51 m

Calculate the extension, ΔL:

- Since the rider is at their maximum height when the rope is unstretched, this means ΔL is the amplitude of the oscillations of the rider
- $h = x + \Delta x + \Delta x + L$
- $2\Delta L = h - x - L = 51 - 3.1 - 23 = 24.9$
- $\Delta L = \frac{24.9}{2} = 12.45$ m *[1 mark]*

Compared forces on the rider in equilibrium:

- When the rider eventually comes to rest, suspended above the ground, their force downwards (weight) will equal the force upwards from Hooke's law
- So, $mg = k\Delta x$ *[1 mark]*

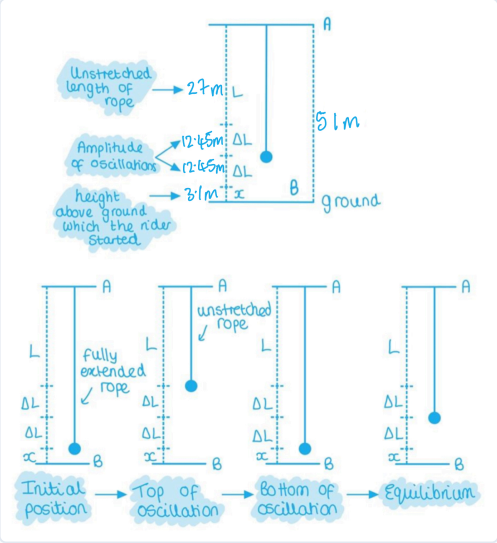
Calculate the spring constant, k:

- $k = \frac{mg}{\Delta x} = \frac{70 \times 9.81}{12.45} = 55.16$ N m^{-1} *[1 mark]*

[Total: 3 marks]

*Although the calculations for this question are not necessarily difficult, the setup of the question requires some thinking. Drawing a free-body diagram can help visualise which forces and distances are required here. The important part is that **A** is adjusted so that the rope is unstretched when the rider reaches the maximum amplitude of their oscillation (i.e. the rope doesn't go completely slack and continue upwards from the force, and instead is still vertically downwards, just unstretched).*

*This means that the combination of the initial height above the ground (x) the maximum (ΔL) and minimum amplitude (ΔL) of the oscillation (which is the extension of the rope due to the rider) **and** the unstretched length of the rope (L) all make up the distance from the ground to point A (h).*



The second half of the diagram shows the movement of the rider from the start to its final position at equilibrium (the rider and the harness is represented by the circle as a point mass. Art is not my strong point). In practice, the rider's oscillations will become smaller and smaller till they reach equilibrium and at rest. This is where their weight downwards is equal to the force upwards due to Hooke's law.

Part (b) [4 marks]

A different ride consists of a 15 m length of rope, stretched into a 'V' shape PQR on a frame, as shown in Fig 1.2. The frame is 16 m wide and the rider stands in the middle at position R. The ends of the elastic rope are fixed to the frame at the point P and Q.

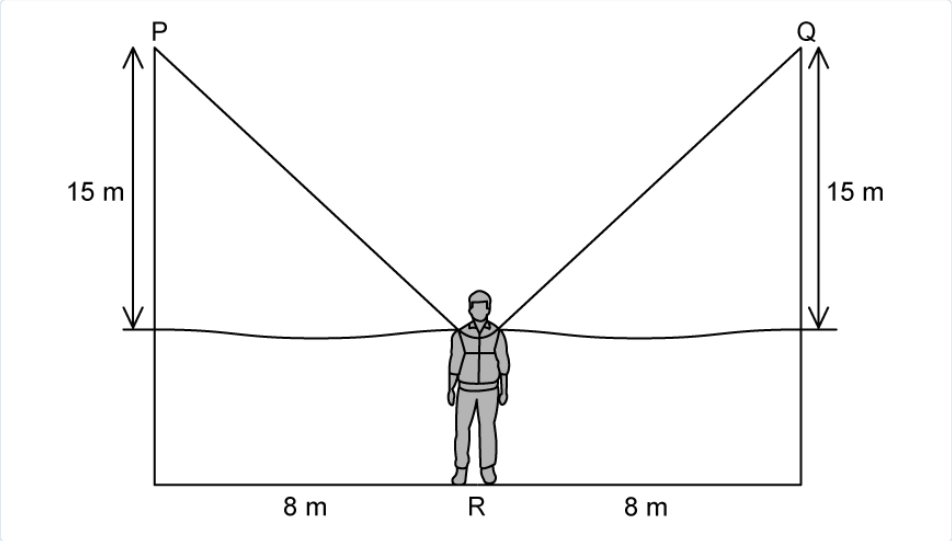


Fig. 1.2

The rider is attached to the midpoint of the elastic rope at R. The Young Modulus of the rope is 1920 Pa, and the cross-sectional area of the rope is 1 m^2.

Calculate the elastic potential energy of the elastic rope in the initial position shown in the diagram.

Mark Scheme (4 marks):

List the known quantities:

- Young Modulus = 1920 N
- Unstretched length of rope, *L* = 15 m

Calculate the extension of the rope:

- The final length of the rope is 2 × PR (or 2 × RQ)
- PR = $\sqrt{8^2 + 15^2}$ = 17 m
- Final length = 2 × 17 = 34 m
- Extension, *x* = 34 – *L* = 34 – 15 = 19 m *[1 mark]*

Calculate the spring constant, *k*:

- Young Modulus = $\frac{FL}{Ax}$
- From Hooke's Law: *F* = *kx*
- Young Modulus = $kx \times \frac{L}{\Delta x} = kL$ *[1 mark]*

- $k = \frac{Young\ modulus}{L} = \frac{1920}{15} = 128$ N m^{-1} *[1 mark]*

Calculate the elastic potential energy, *E*:

- $E = \frac{1}{2}k(\Delta x)^2 = \frac{1}{2} \times 128 \times (19)^2 = 23104 = 23\,000$ J (23 kJ) *[1 mark]*

[Total: 4 marks]

Part (c) **[4 marks]**

The elastic bungee cord is replaced with an elastic which can also provide mechanical resistance when performing fitness exercises. The graph of load against extension for the chord is shown in Fig. 1.3 below.

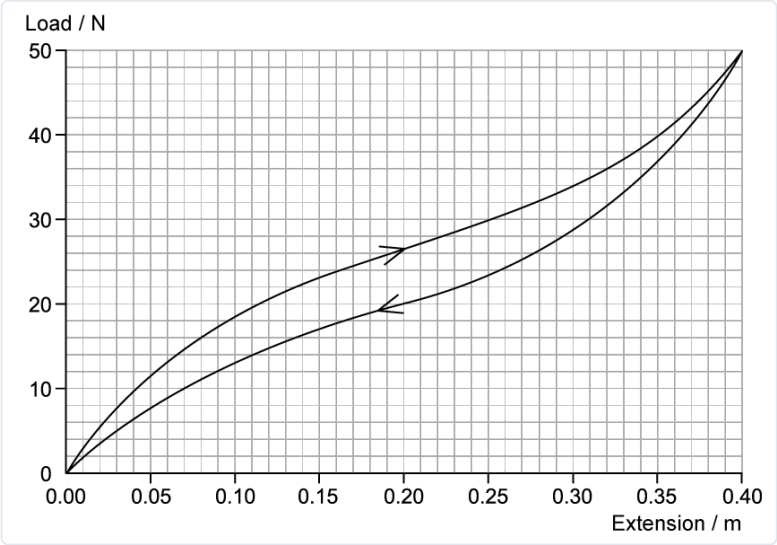


Fig. 1.3

Estimate the minimum energy required to stretch the elastic to an extension of 0.40 m.

Mark Scheme (4 marks):

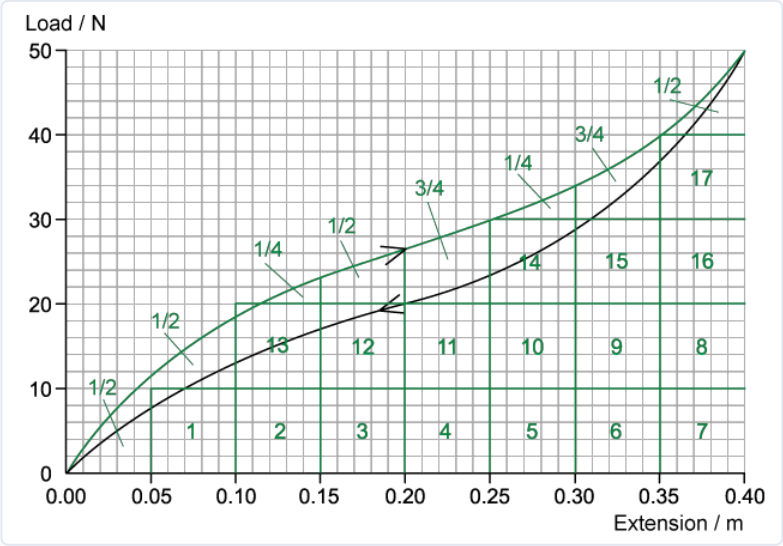
The minimum energy required to stretch the elastic to an extension of 0.40 m is:

- The area under the loading graph

Calculate the value of 1 cm square:

- 1 cm square = 0.05 × 10 = 0.5 J *[1 mark]*

Area under loading graph (work done to stretch the rubber):



Calculate the number of cm squares:

- 17 whole squares + (2 × $\frac{1}{4}$) squares + (4 × $\frac{1}{2}$) squares + (2 × $\frac{3}{4}$) squares = 21 squares
- Total work done to stretch rubber = 0.5 × 21 = 10.5 J (Acceptable answers: 10.25 to 10.75) *[1 mark]*

[Total: 4 marks]

Remember your answer is simply an estimation since everyone will have a different opinion of which squares to count, so don't worry too much about getting the exact value. Always use this method for finding the area under a curved graph.

Remember the area underneath each curve represents slightly different things:

